

FlowForge

**A Cloud-Native Microservices based Task
Management Platform with a Production-
Ready DevSecOps Pipeline**

Noel Mathews

Objective

FlowForge is a cloud-native project and task management platform built on 6 microservices — API Gateway, Auth, Project, Task, Analysis, and a Next.js Frontend — backed by PostgreSQL and Redis.

The Problem: Before DevOps, software delivery was entirely manual. Engineers SSH-ed into servers, copied files by hand, had no monitoring, no automated testing, and no reliable rollback. As applications moved to the cloud and microservices became the standard, this approach completely broke down — you cannot manually manage 6 services, a database, a cache, load balancers, and storage across distributed nodes.

Our Goal: Deploy FlowForge on a production-grade, self-managed Kubernetes cluster on AWS — with three worker nodes, persistent NFS-backed storage, a centralized load balancer, and full namespace isolation between dev and production environments. On top of that infrastructure, build a DevSecOps pipeline that automates every step from code push to cluster deployment — with security scanning, controlled release strategies, deployment governance, and full observability — so the platform runs reliably, safely, and at scale.

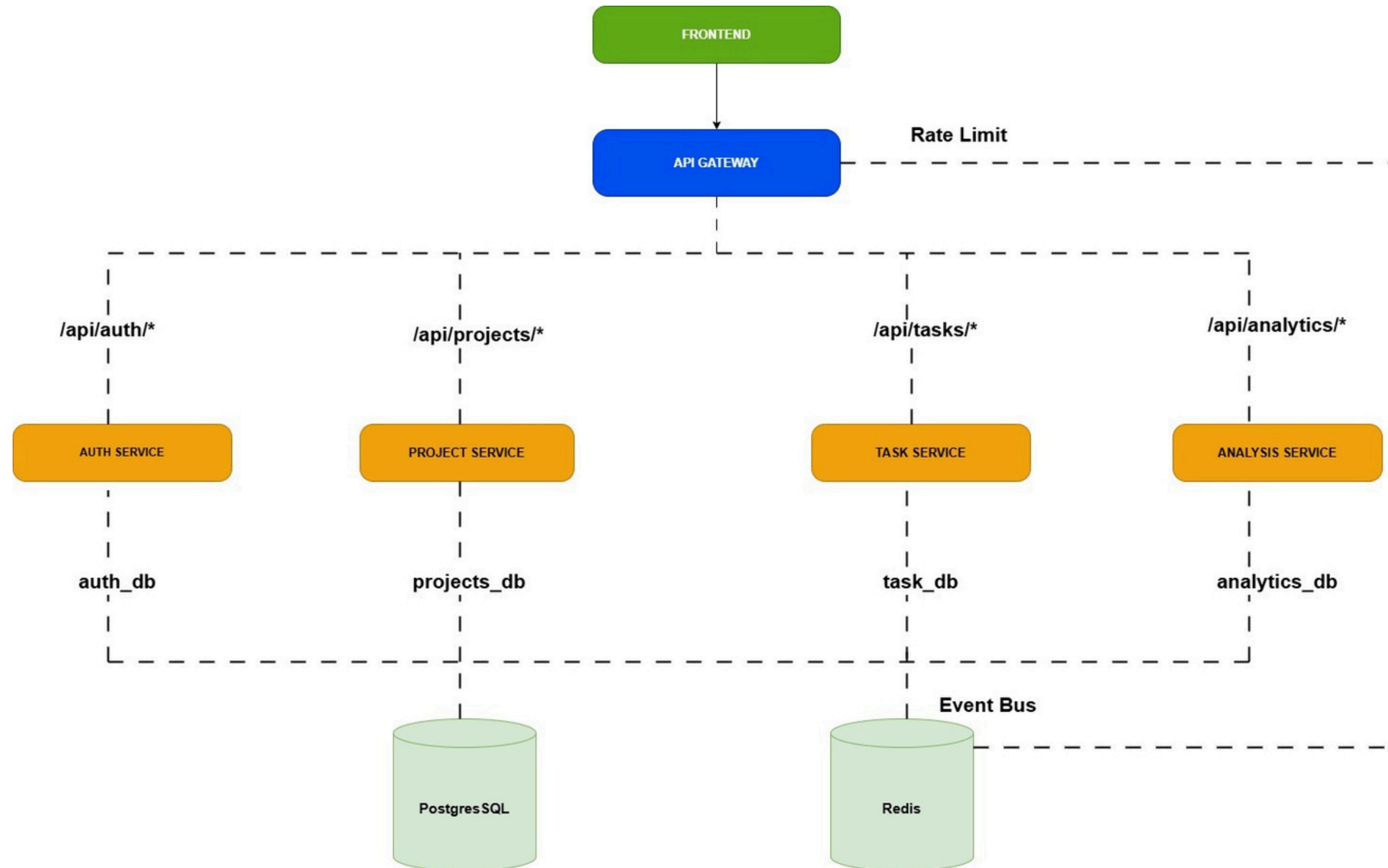
Application Stack

- Frontend: Next.js Framework
- Backend Services: Python FastAPI
- Internal Communication: HTTP REST API calls over ClusterIP with NetworkPolicy
- Data Layer: PostgreSQL and Redis isolated via PVCs

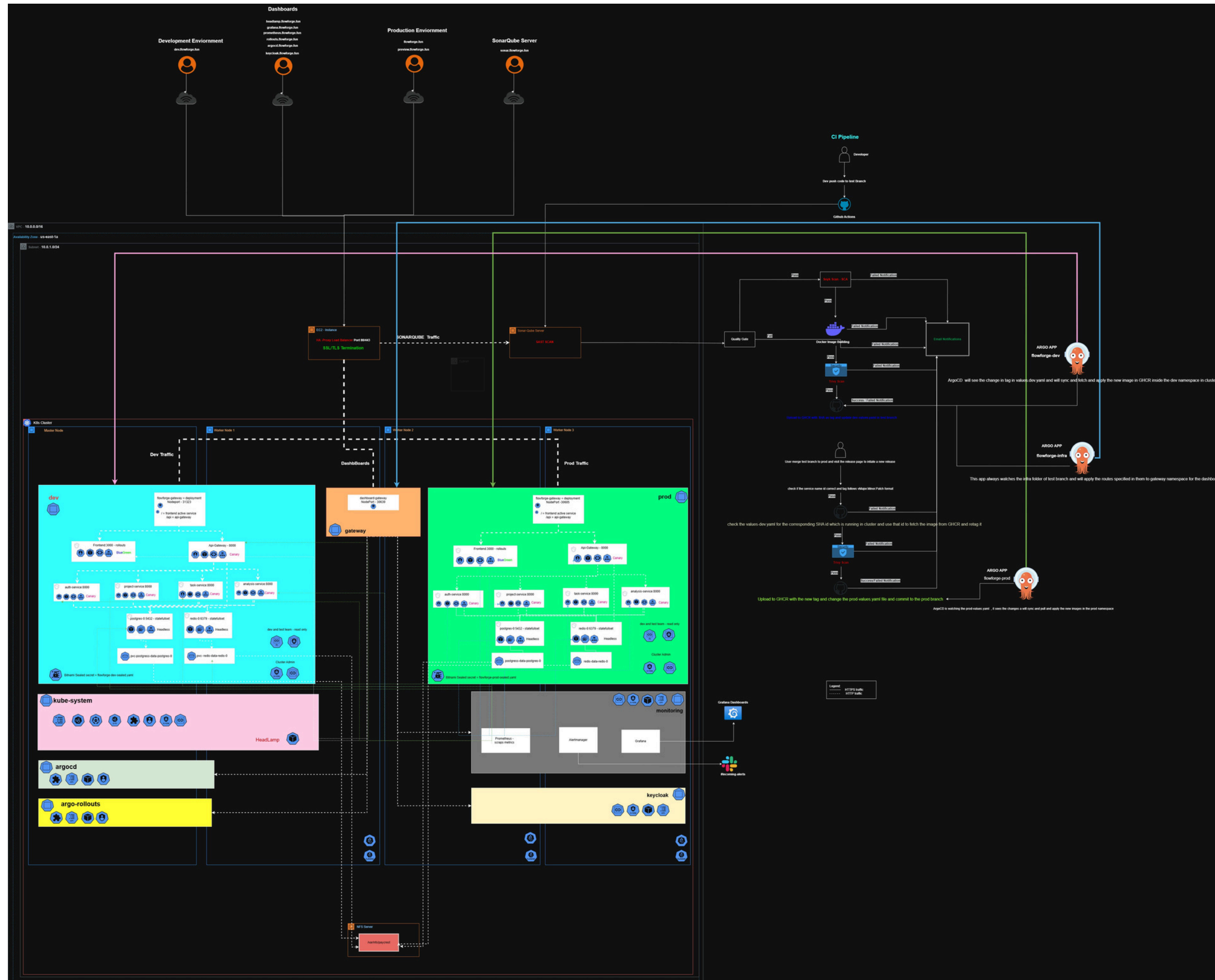
Backend MicroServices:

1. Gateway - Acts as the single entry point for the application, routing incoming client requests to the appropriate backend microservices also handles rate limiting and JWT Validation
2. Auth-service - Handles user registration, authentication, authorization.
3. Project-service - Manages the complete lifecycle, metadata, and configuration of projects created within the system.
4. Task-service - Handles creation, assignment, status tracking, and management of individual tasks within projects.
5. Analysis-service - Processes application data to generate metrics, insights, and analytical reports regarding projects and tasks.

Application Architecture



Devops Architecture with CI CD Pipeline



Services and Ports

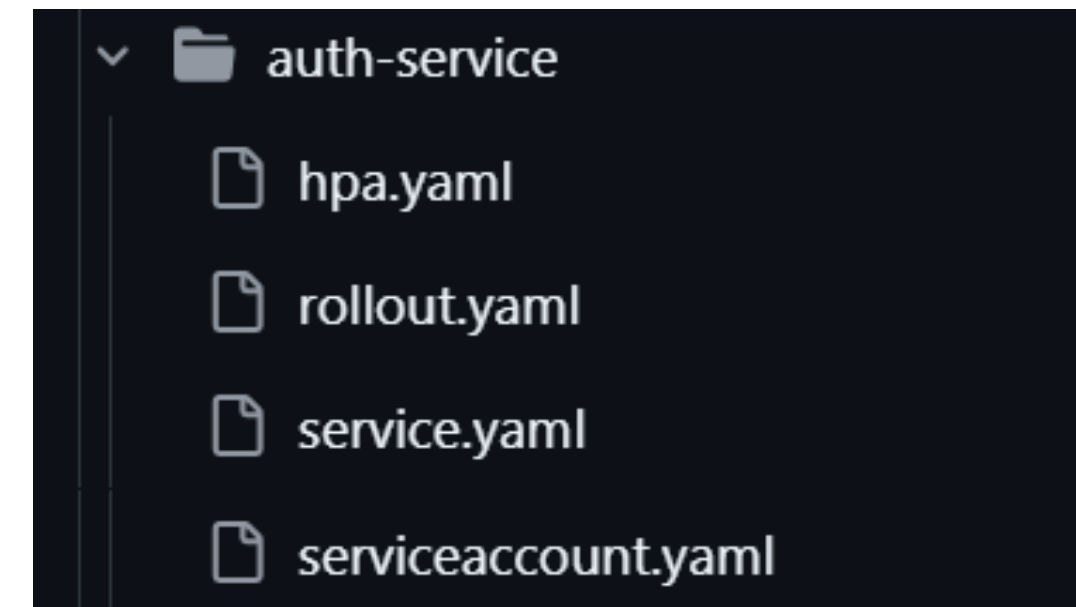
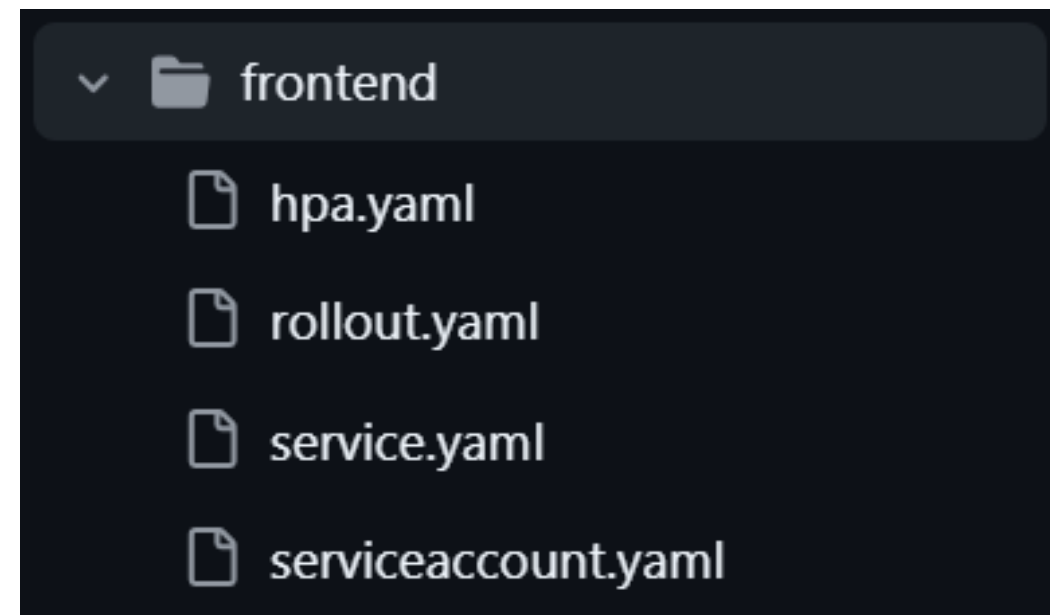
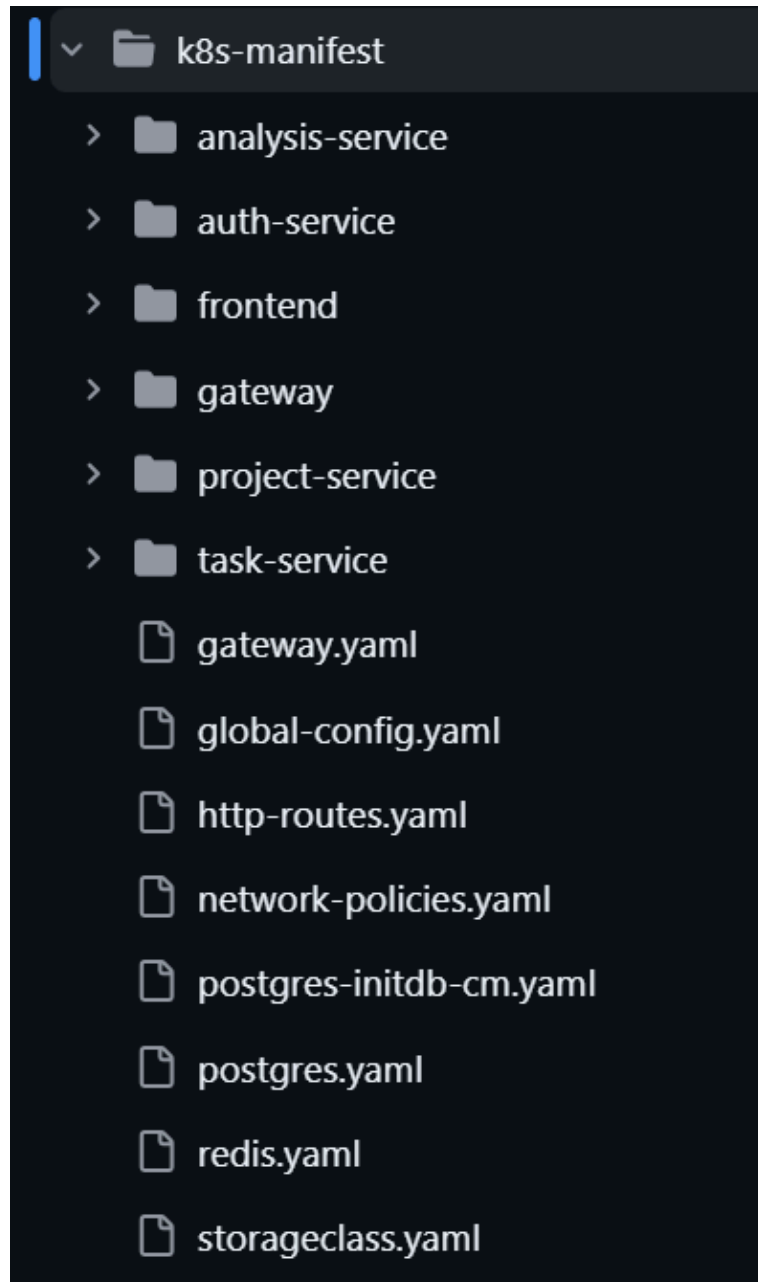
Frontend	3000
Gateway	8000
Auth-Service	8001
Project-Service	8002
Task-Service	8003
Analysis-Service	8004

DockerFiles

```
auth-service > Dockerfile > ...  
1 #auth-service  
2 FROM python:3.11-slim AS builder  
3 WORKDIR /app  
4 COPY requirements.txt .  
5 RUN --mount=type=cache,target=/root/.cache/pip \  
6     pip install --prefix=/install -r requirements.txt  
7  
8 FROM python:3.11-slim AS runner  
9 WORKDIR /app  
10 COPY --from=builder /install /usr/local  
11 RUN addgroup --system stgroup && adduser --system --group stuser  
12 COPY --chown=stuser:stgroup . .  
13 USER stuser  
14 EXPOSE 8001  
15 CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8001"]
```

```
frontend > Dockerfile > ...  
1 FROM node:18-alpine AS deps  
2 RUN apk add --no-cache libc6-compat  
3 WORKDIR /app  
4 COPY package*.json ./  
5 RUN npm ci  
6  
7 FROM node:18-alpine AS builder  
8 WORKDIR /app  
9 ENV NEXT_PUBLIC_API_URL=/api  
10 ENV NEXT_PUBLIC MOCK_MODE=false  
11 COPY --from=deps /app/node_modules ./node_modules  
12 COPY . .  
13 RUN npm run build  
14  
15 FROM node:18-alpine AS runner  
16 WORKDIR /app  
17 ENV NODE_ENV=production  
18 RUN addgroup --system --gid 1001 stgroup \  
19     && adduser --system --uid 1001 --ingroup stgroup stuser  
20 COPY --from=builder /app/public ./public  
21 COPY --from=builder --chown=stuser:stgroup /app/.next/standalone ./  
22 COPY --from=builder --chown=stuser:stgroup /app/.next/static ./next/static  
23 USER stuser  
24 CMD ["node", "server.js"]
```

Kubernetes Manifest Files



Security Aspects

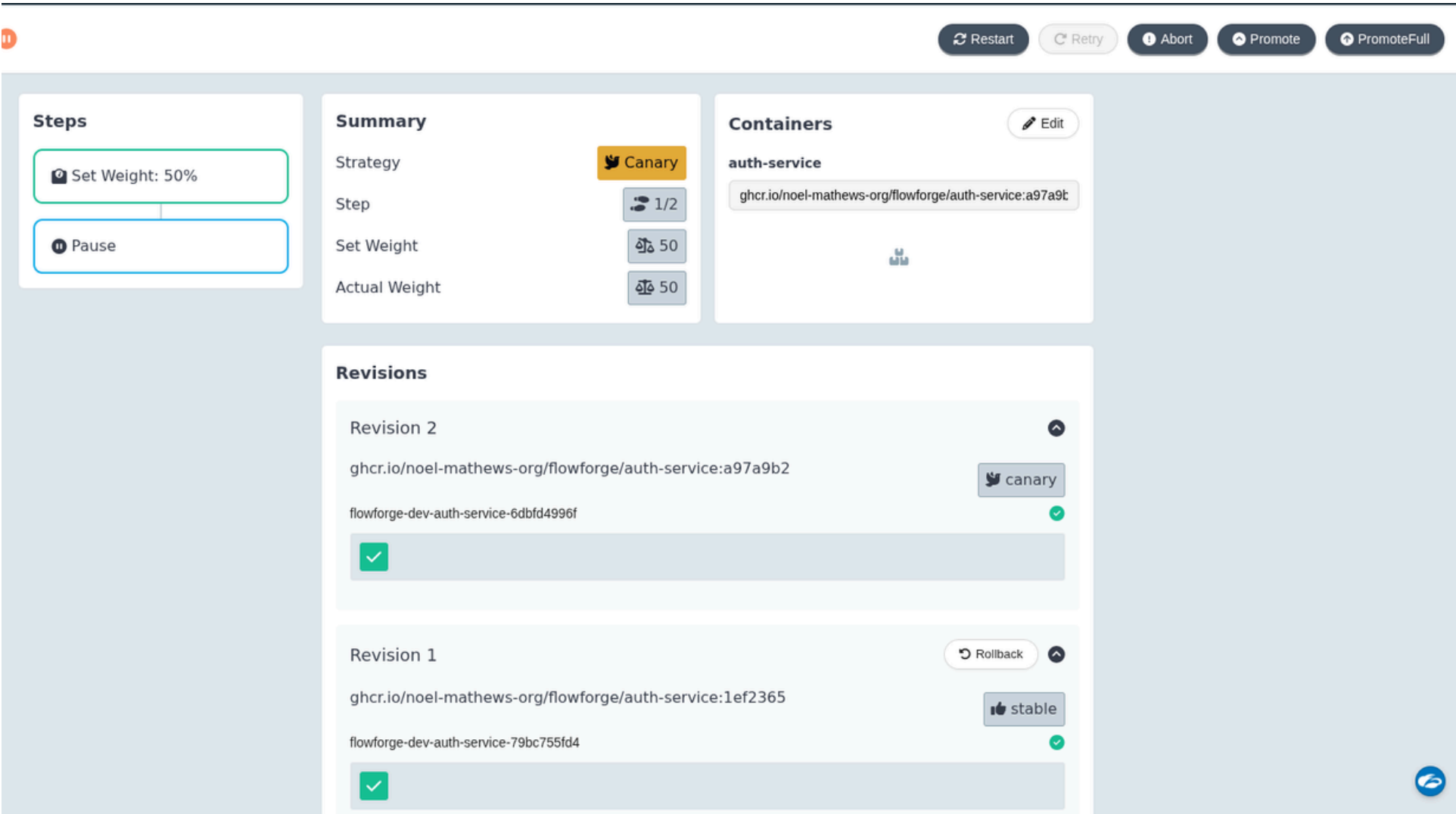
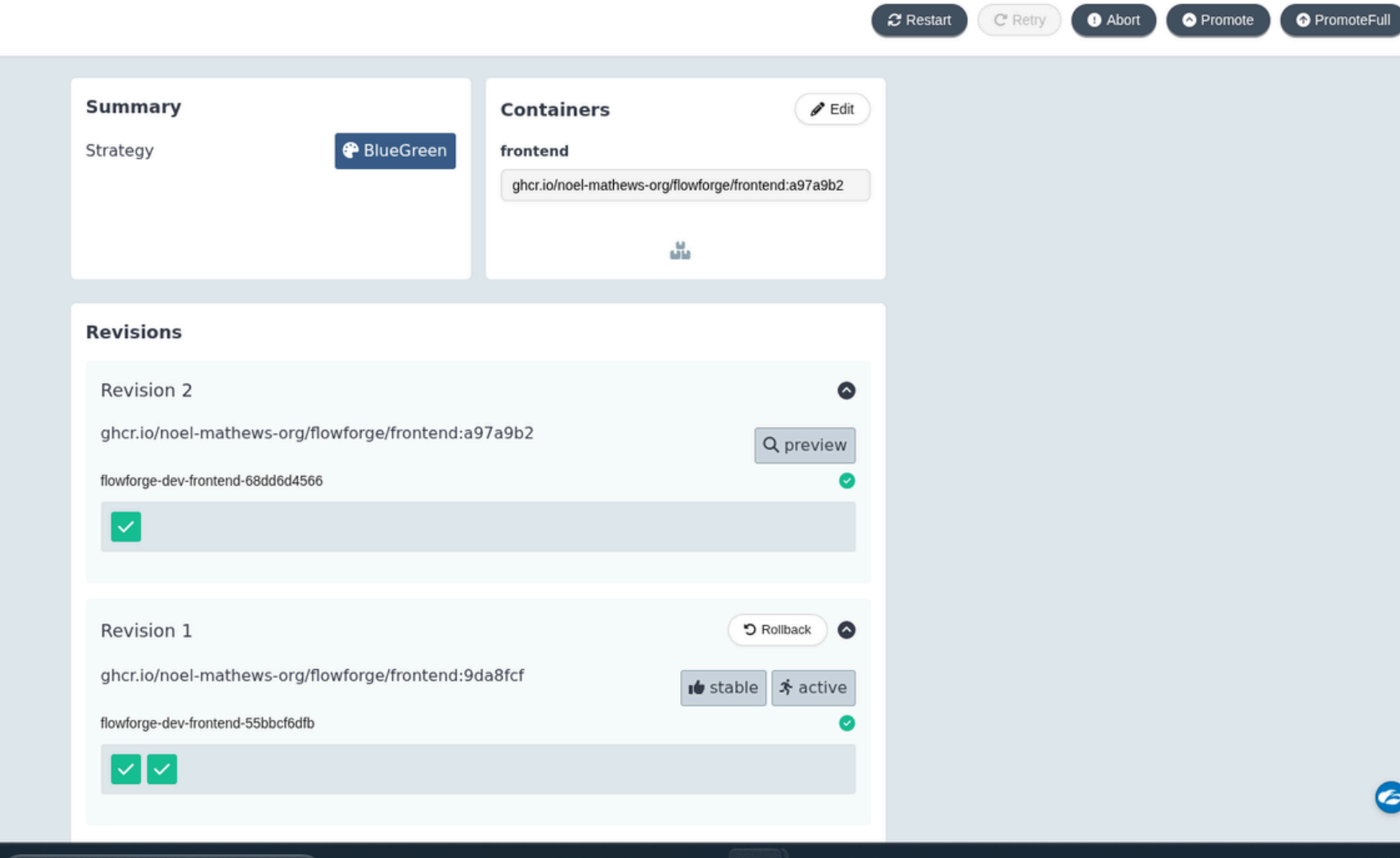
```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: flowforge-dev-auth-service
  namespace: default
automountServiceAccountToken: false
```

Service Accounts - Instead of default SA we have created a SA for each microservice and disabled injecting the token that can be used to communicate with the API Server , and we have given that the containers should run as nonroot only

NetworkPolicy - we have implemented network policies inorder to restrict unauthorized communication in between the components in our namespace and across from other namespaces

Kyverno Cluster Policy - we have implemented cluster policy using kyverno to disallow any new resource creation using latest tag , specified tag should be given .

Deployment Strategies



BlueGreen - Frontend

Microservices - Canary

Helm Charts - GitOps

- ▼  Helm
 - ▼  charts
 - >  analysis-service
 - >  auth-service
 - >  frontend
 - >  gateway
 - >  project-service
 - >  task-service
 - >  templates
 -  Chart.yaml
 -  values-dev.yaml
 -  values-prod.yaml

- ▼  frontend
 - ▼  templates
 -  hpa.yaml
 -  rollout.yaml
 -  service.yaml
 -  serviceaccount.yaml
 -  Chart.yaml
 -  values.yaml

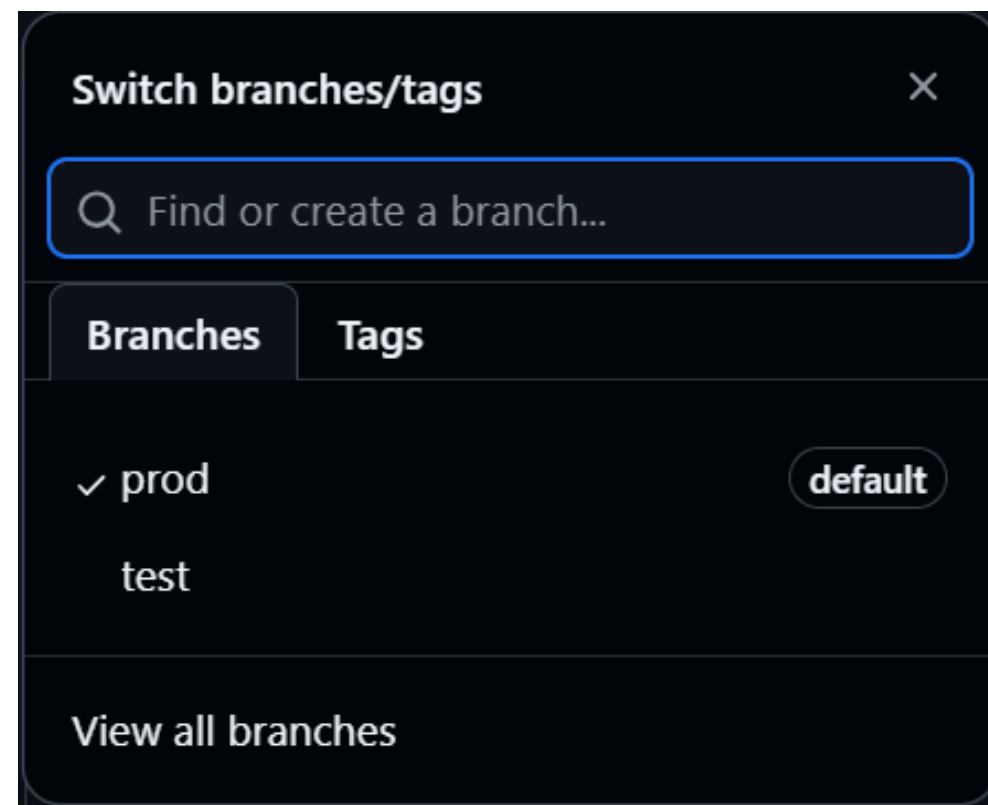
- ▼  auth-service
 - ▼  templates
 -  hpa.yaml
 -  rollout.yaml
 -  service.yaml
 -  serviceaccount.yaml
 -  Chart.yaml
 -  values.yaml

- ▼  infra
 - >  argocd
 - >  infrastructure
 - ▼  kyverno-policies
 -  cluster-policies.yaml
 -  initdb.sql

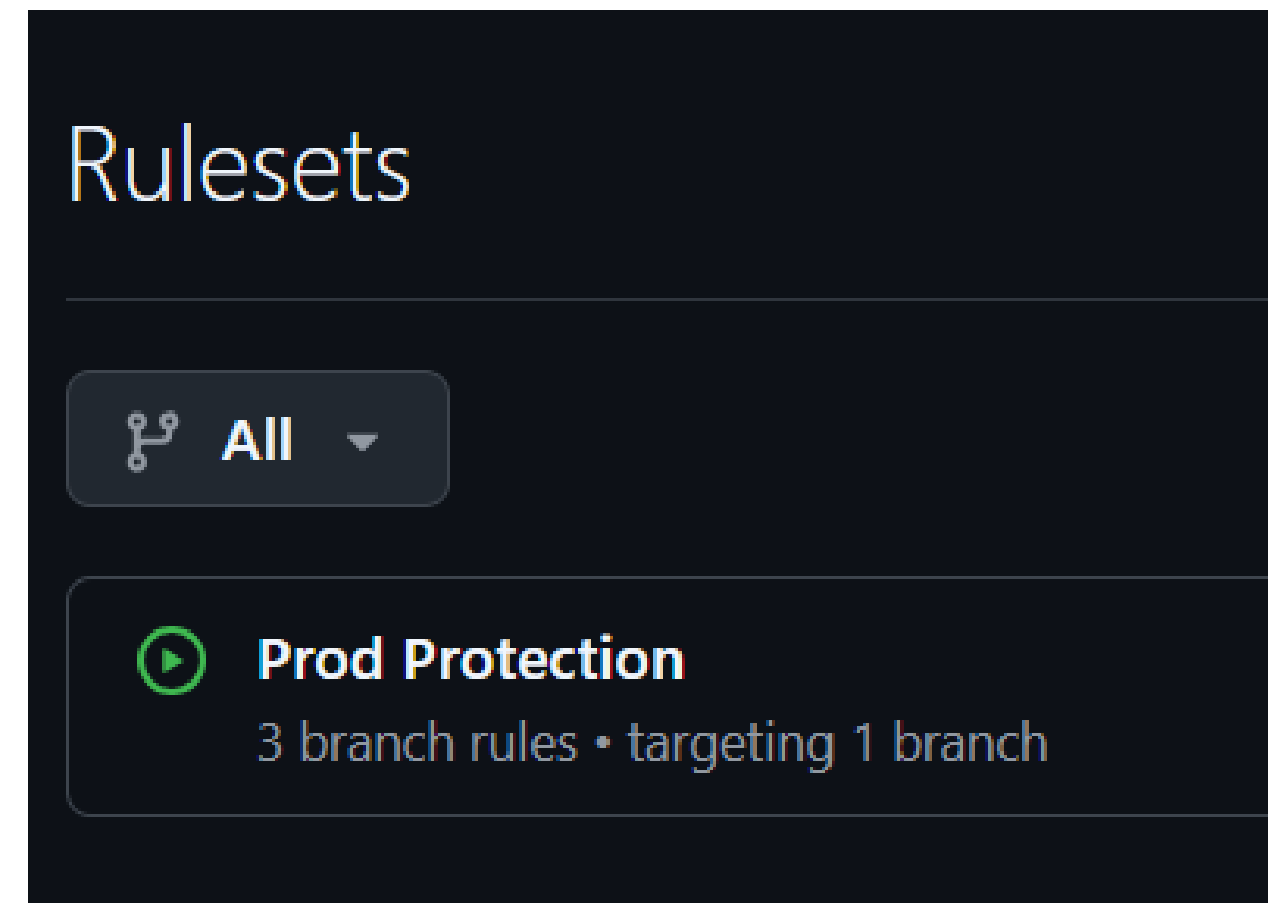
- ▼  infrastructure
 -  alertmanager-sealed.yaml
 -  argocd-route.yaml
 -  dashboard-gateway.yaml
 -  grafana-route.yaml
 -  headlamp-route.yaml
 -  keycloak-route.yaml
 -  keycloak-setup.yaml
 -  loki-route.yaml
 -  monitoring-rules.yaml
 -  prometheus-route.yaml
 -  rollouts-dashboard.yaml
 -  rollouts-route.yaml

Branching Strategy

Trunk Based



Branch Protection



- No direct pushes to prod branch
- Pull request required before merge
- At least one approval required
- Status checks must pass before merge

Secret Management

Repository secrets

[New repository secret](#)

Name 	Last updated	
 SONAR_TOKEN_ANALYSIS	2 days ago	 
 SONAR_TOKEN_AUTH	2 days ago	 
 SONAR_TOKEN_FRONTEND	2 days ago	 
 SONAR_TOKEN_GATEWAY	2 days ago	 
 SONAR_TOKEN_PROJECT	2 days ago	 
 SONAR_TOKEN_TASK	2 days ago	 

Organization secrets

[Manage organization secrets](#)

Name 	Last updated	
 DEVELOPMENT_TEAM_EMAIL	2 days ago	
 GH_PAT	2 days ago	
 MAIL_PASSWORD	2 days ago	
 MAIL_USERNAME	2 days ago	
 SNYK_TOKEN	2 days ago	
 SONAR_HOST_URL	2 days ago	

```
apiVersion: bitnami.com/v1alpha1
kind: SealedSecret
metadata:
  name: flowforge-secret
  namespace: prod
spec:
  encryptedData:
    DATABASE_PASSWORD: "AgAcWW/n3C0YANQt8s4DI dy21NEcOkxCHLHH+BLGEp65TnJEx/tyGz7dQWhZX1aqG+hQ8kqTx6r0
    JWT_SECRET: "AgAgVJ94zcxKInpPQqimgviKIi+E1yVi/Wt+qF+0t74dcN3FsytII7+hQH6BI2uTpThiwTQHgxZE/yGEbvg
    SMTP_PASSWORD: "AgBLaorEt1Gy5KcbIIrd0qw670yhQ260kARfCGJWTzbU1+BRPV4x7oduTPV8MKjjJr+uG71LyYFyKoz5
  template:
    metadata:
      name: flowforge-secret
      namespace: prod
```

```
apiVersion: bitnami.com/v1alpha1
kind: SealedSecret
metadata:
  name: flowforge-secret
  namespace: dev
spec:
  encryptedData:
    DATABASE_PASSWORD: "AgCDJ/V/Y3EEyYEwJ+5DUfuQyhZuLV0E9tdoZkA1sFqR9p72rKzFIuIok8IYDJwxjEbn
    JWT_SECRET: "AgBNWStK151wOF62YuC21gnfE2nZi/t+QfjgqapYcnNNLmTsINfg83YzIr8W0zDaLEsr1TdzUf1
    SMTP_PASSWORD: "AgDFrLheIKB80/SZVfN01pk8DNWhF9+vQA9FhBq+I/saLtkH4G7RI1UFcgwLKxnBYAqQ0Lbj
  template:
    metadata:
      name: flowforge-secret
      namespace: dev
```


CI-CD Pipeline

run-ci / build-and-deploy

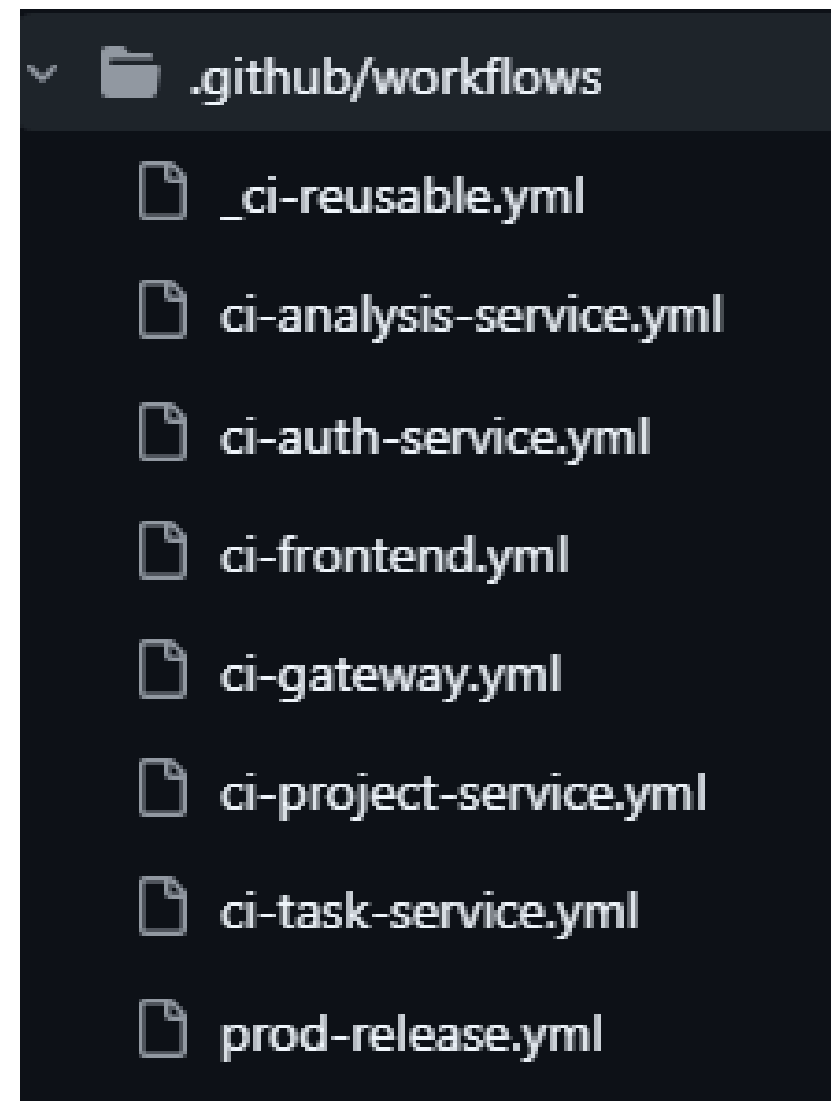
succeeded yesterday in 1m 32s

- > Build sonarsource/sonarqube-scan-action@v2
- > Run actions/checkout@v4
- > Set Variables
- > SonarQube Scan
- > SonarQube Quality Gate
- > Setup Python
- > **Snyk Scan (Python)**
- Setup Node
- Snyk Scan (Frontend)
- > Lowercase Repository Name
- > Build Docker Image
- > Trivy Scan
- > Login to GHCR
- > Push to GHCR
- > Update values-dev.yml and Push
- > Post Login to GHCR
- > Post Trivy Scan
- > Post Setup Python
- > Post SonarQube Scan
- > Post Run actions/checkout@v4
- > Complete job

promote-to-prod

succeeded 10 hours ago in 22s

- > Set up job
- > Run actions/checkout@v4
- > Parse Release Tag
- > Fetch Dev SHA
- > Lowercase Repository Name
- > Login to GHCR
- > Pull Image
- > Trivy Scan Before Promotion
- > Retag with SemVer and Push
- > Update values-prod.yml and Push
- > Post Trivy Scan Before Promotion
- > Post Login to GHCR
- > Post Run actions/checkout@v4
- > Complete job



Shift Left Security - Securing Left side before moving to right side

Reuseable Workflows - We dont need to create seperate pipeline for each microservice , we can use workflow call to call this reuseable workflow by path based triggers

Deployment Governance - we use 7 letters of the commit SHA of the developer to track which image is created and is responsible by whom

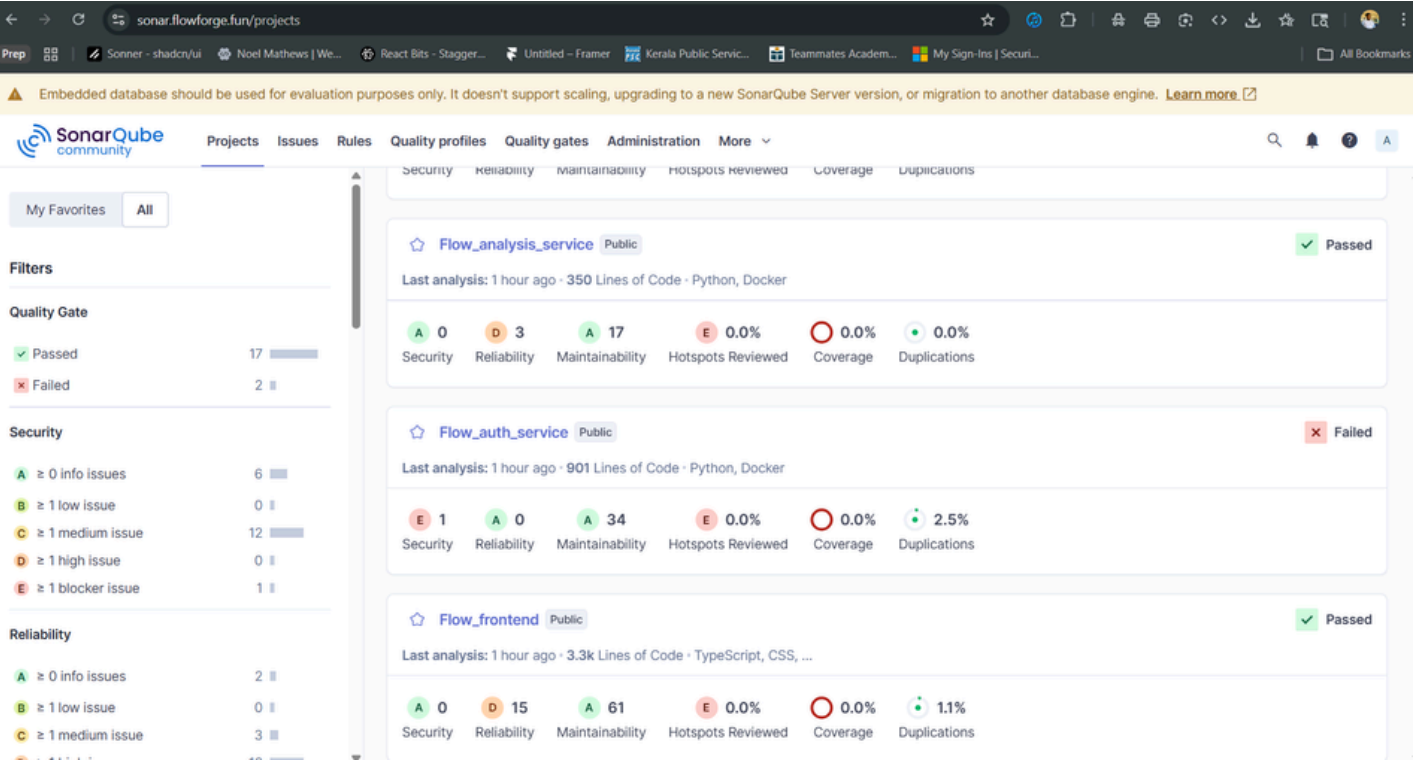
Immutable Promotion - we do not rebuild the image that wa salready tested and is ready for production , we just retag it and deploy to production which helps us to maintain the integrity of the image

Shift Left Security

```
154 Issues to fix by upgrading dependencies:
155
156 Upgrade python-jose@3.3.0 to python-jose@3.4.0 to fix
157 X Improper Verification of Cryptographic Signature [High Severity][https://nvd.nist.gov/vuln/detail/CVE-2025-69720]
158 introduced by python-jose@3.3.0
159 X Improper Handling of Highly Compressed Data (Data Amplification) [High Severity][https://nvd.nist.gov/vuln/detail/CVE-2025-69720]
160 introduced by python-jose@3.3.0
161
162 Pin starlette@0.41.3 to starlette@0.49.1 to fix
163 X Regular Expression Denial of Service (ReDoS) [High Severity][https://nvd.nist.gov/vuln/detail/CVE-2026-29111]
164 introduced by fastapi@0.115.5 > starlette@0.41.3
165
166
167 Issues with no direct upgrade or patch:
168 X Timing Attack [High Severity][https://security.snyk.io/vuln/SNYK-PYTHON-ECDH-2025-69720]
169 introduced by python-jose@3.3.0 > ecdsa@0.19.2
170 No upgrade or patch available
171 X Missing Encryption of Sensitive Data [High Severity][https://security.snyk.io/vuln/SNYK-PYTHON-ECDH-2025-69720]
172 introduced by python-jose@3.3.0 > ecdsa@0.19.2
173 No upgrade or patch available
174
175
176
177 Organization:    noelmc-ust
178 Package manager: pip
179 Target file:     requirements.txt
180 Project name:    task-service
181 Open source:     no
182 Project path:    /home/runner/work/FlowForge/FlowForge/task-service
183 Licenses:        enabled
184
185
186 Error: Process completed with exit code 1.
```

```
376 ghcr.io/noel-mathews-org/flowforge/auth-service:3ceda5f (debian 13.4)
377
378 =====
379 Total: 6 (HIGH: 6, CRITICAL: 0)
380
381
382 | Library | Vulnerability | Severity | Status | Installed Version | Fixed Version | Title |
383 |-----|-----|-----|-----|-----|-----|-----|
384 | libncursesw6 | CVE-2025-69720 | HIGH | affected | 6.5+20250216-2 | | ncurses: ncurses: Buffer overflow vulnerability may lead to |
385 | | | | | | | arbitrary code execution. |
386 | | | | | | | https://avd.aquasec.com/nvd/cve-2025-69720 |
387 |-----|-----|-----|-----|-----|-----|-----|
388 | libsystemd0 | CVE-2026-29111 | | | 257.9-1~deb13u1 | | systemd: systemd: Arbitrary code execution or Denial of |
389 | | | | | | | Service via spurious IPC... |
390 | | | | | | | https://avd.aquasec.com/nvd/cve-2026-29111 |
391 |-----|-----|-----|-----|-----|-----|-----|
392 | libtinfo6 | CVE-2025-69720 | | | 6.5+20250216-2 | | ncurses: ncurses: Buffer overflow vulnerability may lead to |
393 | | | | | | | arbitrary code execution. |
394 | | | | | | | https://avd.aquasec.com/nvd/cve-2025-69720 |
395 |-----|-----|-----|-----|-----|-----|-----|
396 | libudev1 | CVE-2026-29111 | | | 257.9-1~deb13u1 | | systemd: systemd: Arbitrary code execution or Denial of |
397 | | | | | | | Service via spurious IPC... |
398 | | | | | | | https://avd.aquasec.com/nvd/cve-2026-29111 |
399 |-----|-----|-----|-----|-----|-----|-----|
400 | ncurses-base | CVE-2025-69720 | | | 6.5+20250216-2 | | ncurses: ncurses: Buffer overflow vulnerability may lead to |
401 | | | | | | | arbitrary code execution. |
402 | | | | | | | https://avd.aquasec.com/nvd/cve-2025-69720 |
403 |-----|-----|-----|-----|-----|-----|-----|
404 | ncurses-bin | | | | | | |
405 | | | | | | |
406 | | | | | | |
407 |-----|-----|-----|-----|-----|-----|-----|
408
```

```
> Run actions/checkout@v4
> Set Variables
> SonarQube Scan
v SonarQube Quality Gate
  1 ▶ Run sonarsource/sonarqube-quality-gate-action@v1
  6 ▶ Run $GITHUB_ACTION_PATH/script/check-quality-gate.sh .scannerwork/report-
 11 ..
 12 X Quality Gate has FAILED.
 13 Error: Process completed with exit code 1.
Setup Python
```



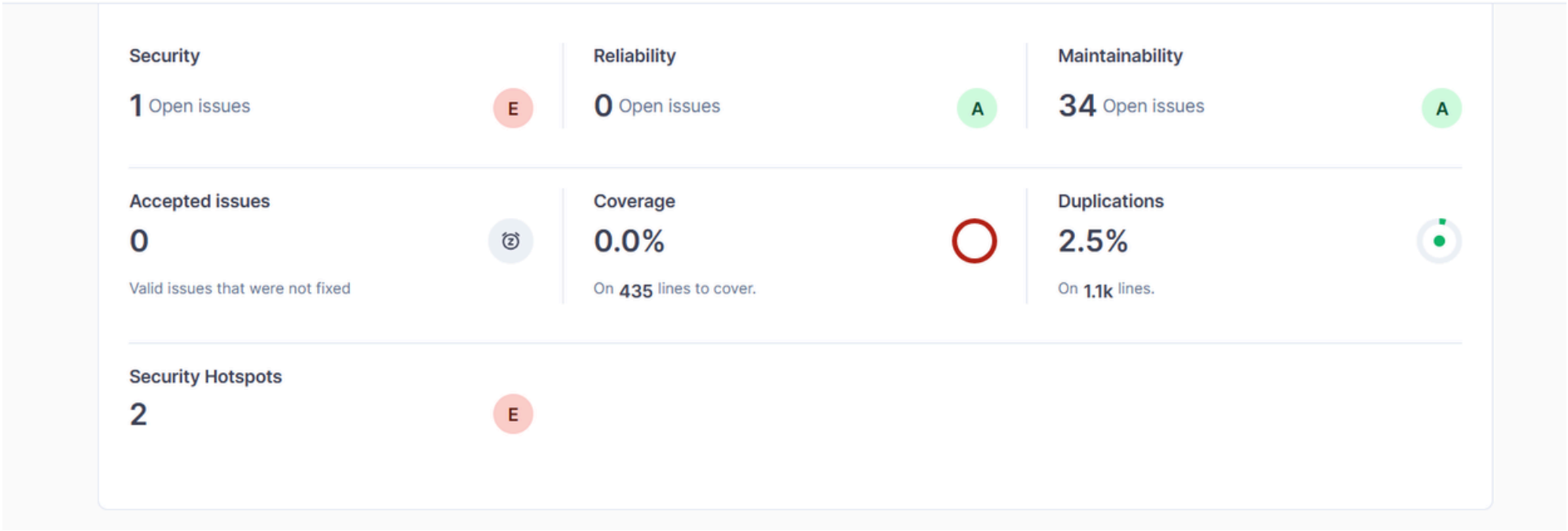
SonarQube Quality Gate - Help us to make sure our code maintains a certain level of benchmark for Reliability , Securty and Maintainability

Flow_auth_service / Overview

Overview

Set as homepage

901 Lines of Code • Version not provided • Last analysis 1 hour ago



Manual Gates

AgentsActionsProjectsWikiSecurity and qualityInsightsSettings

ReleasesTagsDraft a new releaseFind a release

1 hour ago

Noel9812

task-service-v...

a77153a

Compare

Task-service-v1.1.2

task-service-v1.1.2

chore(release): promote auth-service to v1.1.2 in PROD

Assets 2

1 hour ago

Noel9812

project-service...

a77153a

Compare

Project-service-v1.1.2

project-service-v1.1.2

chore(release): promote auth-service to v1.1.2 in PROD

Assets 2

Source code (zip)1 hour ago

Source code (tar.gz)1 hour ago

Releases

New release

Tag: frontend-v1.1.2Target: prod

Excellent! This tag will be created from the target when you publish this release.

Release title

Frontend v1.1.2

Release notes

Select a previous tag to create generated release notes

Previous tag: AutoGenerate release notes

WritePreview

Production Release of Frontend

Version - v1.1.2

Markdown is supported

Paste, drop, or click to add files

Attach binaries by dropping them here or selecti

All workflows

Showing runs from all workflows

78 workflow runs

Frontend v1.1.2

Production Release Pipeline #23: Release frontend-v1.1.2 published by Noel98

Jobs and Notifications

ci-auth-service.yml

on: push



prod-release.yml

on: release



FlowForge DevOps <bloodymaryy77@gmail.com>
to me ▼

Sun, Apr 26, 4:13 PM (1

CI Pipeline PASSED
push on test | Service: task-service

COMMIT DETAILS

Author	Noel Mathews
Commit ID	a97a9b2
Message	Demonstrating the Deployment and Pipeline

PIPELINE RESULTS

View Logs

FlowForge DevOps <bloodymaryy77@gmail.com>
to me ▼

Sun, Apr 26, 4:13 PM (12 hours a

CI Pipeline FAILED
push on test | Service: project-service

COMMIT DETAILS

Author	Noel Mathews
Commit ID	a97a9b2
Message	Demonstrating the Deployment and Pipeline

PIPELINE RESULTS

View Logs



FlowForge DevOps <bloodymaryy77@gmail.com>
to me ▾

Sun, Apr 26, 2:32 AM (1 c

✖ Release Pipeline FAILED

release on project-service-v1.1.1

RELEASE DETAILS

Author **Noel Mathews**
Commit ID [e88633b](#)
Tag *project-service-v1.1.1*

PIPELINE RESULTS

View Logs



FlowForge DevOps <bloodymaryy77@gmail.com>
to me ▾

Sun, Apr 26, 2:42 AM (1 day

✔ Release Pipeline PASSED

release on frontend-v1.1.1

RELEASE DETAILS

Author **FlowForge Release Bot**
Commit ID [3e40aad](#)
Tag *frontend-v1.1.1*

PIPELINE RESULTS

View Logs

Argo Rollouts Dashboard



argo

Rollouts v1.2.0+c40cb0d

0

1

0

0

6

0

0

0

Filter by name or label tag:value

☆

flowforge-dev-analysis-service

Strategy

Canary

Weight

50

flowforge-dev-analysis-ser...b9c686454

✓

Revision 2

✓

flowforge-dev-analysis-ser...84cdd9c5cd

✓

Revision 1

✓

Restart

Promote

☆

flowforge-dev-auth-service

Strategy

Canary

Weight

50

flowforge-dev-auth-service-6dbfd4996f

✓

Revision 2

✓

flowforge-dev-auth-service-79bc755fd4

✓

Revision 1

✓

Restart

Promote

☆

flowforge-dev-project-service

Strategy

Canary

Weight

50

flowforge-dev-project-servi...5dfcc647d8

✓

Revision 2

☆

flowforge-dev-task-service

Strategy

Canary

Weight

50

flowforge-dev-task-service-6dcf9588c7

✓

Revision 2

☆

flowforge-dev-frontend

Strategy

BlueGreen

flowforge-dev-frontend-68dd6d4566

✓

Revision 2

✓

flowforge-dev-frontend-55bbcf6dfb

✓

Revision 1

✓

BlueGreenPause

Restart

Promote

☆

flowforge-dev-gateway

Strategy

Canary

Weight

50

flowforge-dev-gateway-589b549c9

✓

Revision 2

✓

flowforge-dev-gateway-85d9b4bb47

✓

Revision 1

✓

Restart

Promote

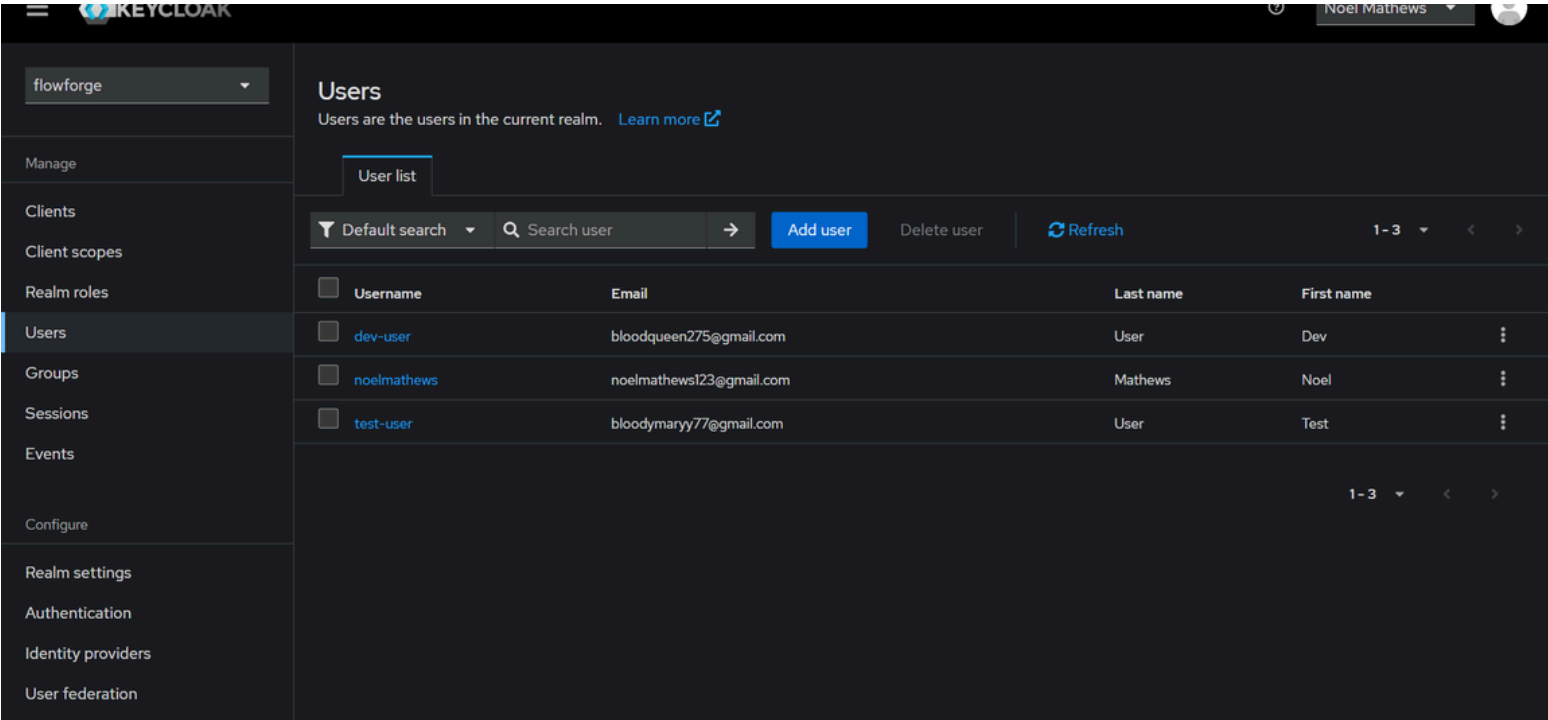
6

Access Control and Team Governance

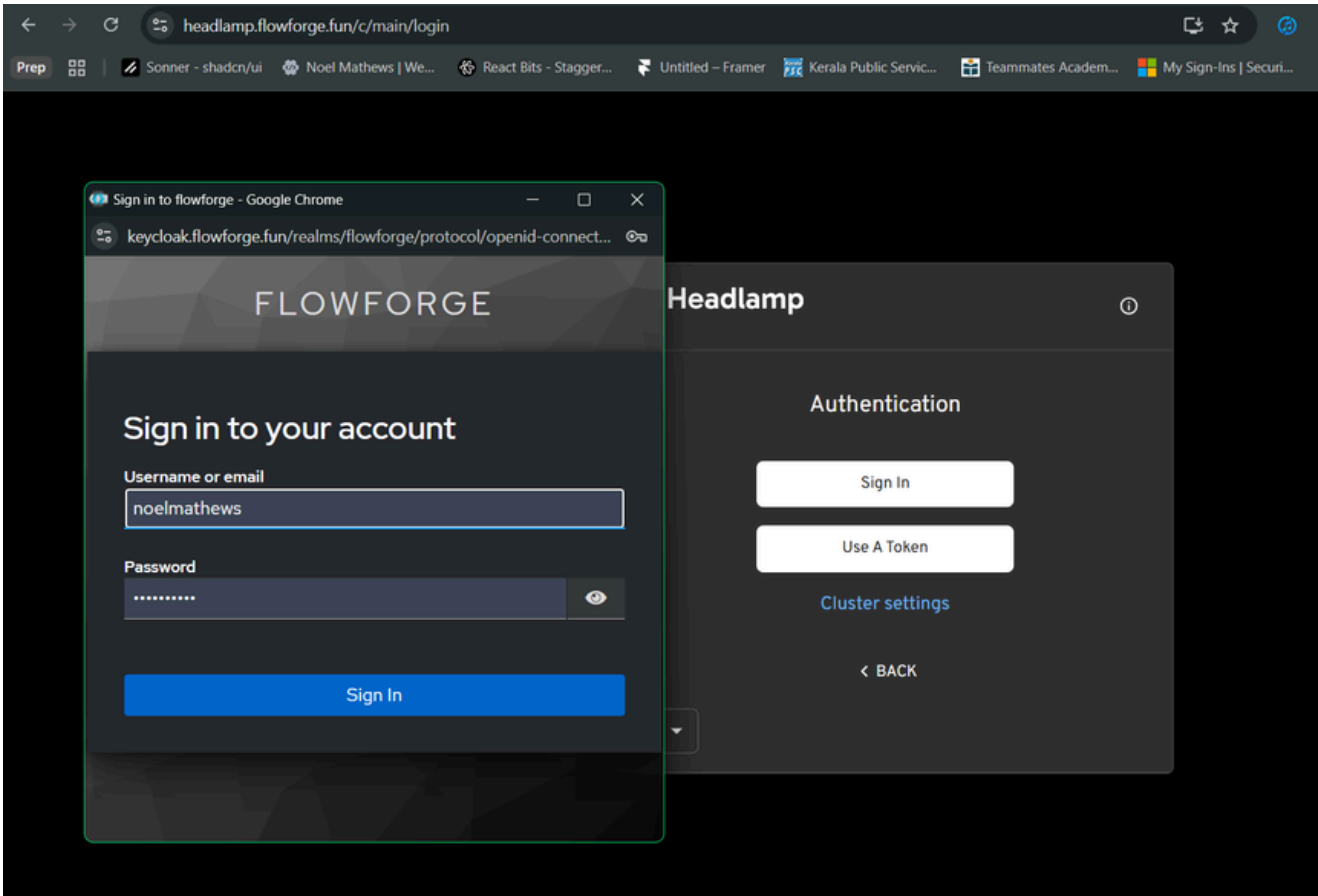
```
root@ip-10-0-1-136:/home/ubuntu# echo "export KUBECONFIG=$(pwd)/flowforge.yaml" >> ~/.bashrc
root@ip-10-0-1-136:/home/ubuntu# source ~/.bashrc
root@ip-10-0-1-136:/home/ubuntu# kubectl get pods -n dev
Username: noelmathews
Password:
NAME                                READY   STATUS    RESTARTS   AGE
flowforge-dev-analysis-service-b9c686454-ctdc7   1/1     Running   1 (30m ago)   8h
flowforge-dev-analysis-service-b9c686454-x4mmm   1/1     Running   1 (30m ago)   8h
flowforge-dev-auth-service-6dbfd4996f-kqw5v      1/1     Running   1 (28m ago)   8h
flowforge-dev-auth-service-6dbfd4996f-rcxt5      1/1     Running   2 (27m ago)   8h
flowforge-dev-frontend-68dd6d4566-nkq4k         1/1     Running   1 (28m ago)   8h
```

Keycloak

```
root@ip-10-0-1-136:/home/ubuntu# kubectl get po -n pod
Username: dev-user
Password:
Error from server (Forbidden): pods is forbidden: User "bloodqueen275@gmail.com" cannot list pods in the namespace "pod"
root@ip-10-0-1-136:/home/ubuntu#
```



HeadLamp - we used keycloak as a signin provider for our team memebers to access the cluster via headlamp



Headlamp

Search

Press / to search

Clusters

main

Namespaces

Nodes

Advanced Search (Beta)

Map

Workloads

Storage

Network

Gateway (beta)

Security

Configuration

Custom Resources

Create

v1.34.0

Overview

CPU Usage

0.67 / 8 units

8.4 %

Memory Usage

8.74 / 15.27 GB

57.3 %

Pods

90 / 90

Requested

100.0 %

Nodes

4 / 4 Ready

100.0 %

Events

Only warnings (2)

Namespaces

prod, dev

Type	Name	Namespace	Reason	Message	Count	Last Seen	Actions
Pod	flowforge-prod-frontend-899d46654-kcqbz	prod	Killing	Stopping container frontend	1	24m	
Pod	flowforge-prod-frontend-899d46654-ncp8v	prod	Killing	Stopping container frontend	1	24m	
ReplicaSet	flowforge-prod-frontend-899d46654	prod	SuccessfulDelete	Deleted pod: flowforge-prod-frontend-899d46654-kcqbz	1	24m	
ReplicaSet	flowforge-prod-frontend-899d46654	prod	SuccessfulDelete	Deleted pod: flowforge-prod-frontend-899d46654-ncp8v	1	24m	
Rollout	flowforge-prod-frontend	prod	ScalingReplicaSet	Scaled down ReplicaSet flowforge-prod-frontend-899d46654 (revision 3) from 2 to 0	1	24m	

Prometheus and Alert Manager

Alerts we setup for our project

 Prometheus

Query

Alerts

Status

Filter by rule group state

Filter by rule name or labels

<

1

2

3

>

flowforge-application-alerts

/etc/prometheus/rules/prometheus-prometheus-kube-prometheus-prometheus-rulefiles-0/monitoring-flowforge-alerts-9b1b040c-c659-4c43-9558-fc3bd170009c.yaml

INACTIVE (2)

ServiceDown

PodCrashLooping

flowforge-infrastructure-alerts

/etc/prometheus/rules/prometheus-prometheus-kube-prometheus-prometheus-rulefiles-0/monitoring-flowforge-alerts-9b1b040c-c659-4c43-9558-fc3bd170009c.yaml

INACTIVE (1)

NodeDown

network-and-api

/etc/prometheus/rules/prometheus-prometheus-kube-prometheus-prometheus-rulefiles-0/monitoring-infrastructure-alerts-a1db3520-2d61-416b-89ee-f48ee0c9791b.yaml

PENDING (2)

INACTIVE (3)

KubeAPILatency

CoreDNSErrors

IngressHighErrorRate

N

bloodymaryy77@gmail.com

to me

[FIRING:1] KubeAPILatency (monitoring/prometheus-kube-prometheus-prometheus warning)

Inbox x

12:18 AM (4 hours ago)

☆

😊

1 alert for alertname=KubeAPILatency

View In Alertmanager

[1] Firing

Labels

alertname = KubeAPILatency

prometheus = monitoring/prometheus-kube-prometheus-prometheus

severity = warning

Annotations

description = 99th percentile API latency is over 1s.

[Source](#)

☆

all-devops-alerts

👤 Invite team

💬 Messages

📄 Add canvas

+



FlowForge-Monitor

APP

3:24 AM

[FIRING:1] KubePodCrashLooping prometheus-kube-state-metrics (redis http 192.168.0.240:8080 kube-state-metrics prod redis-0 monitoring/prometheus-kube-prometheus-prometheus CrashLoopBackOff warning 0279e7f0-bd8f-4e13-82db-f8bb0285dd68)

[FIRING:2] KubePodCrashLooping prometheus-kube-state-metrics (redis http 192.168.0.240:8080 kube-state-metrics redis-0 monitoring/prometheus-kube-prometheus-prometheus CrashLoopBackOff warning)

3:32

[RESOLVED] NodeHighNumberConntrackEntriesUsed prometheus-prometheus-node-exporter (node-exporter http-metrics 10.0.1.155:9100 node-exporter monitoring prometheus-prometheus-node-exporter-49dh9 monitoring/prometheus-kube-prometheus-prometheus warning)



FlowForge-Monitor

APP

3:39 AM

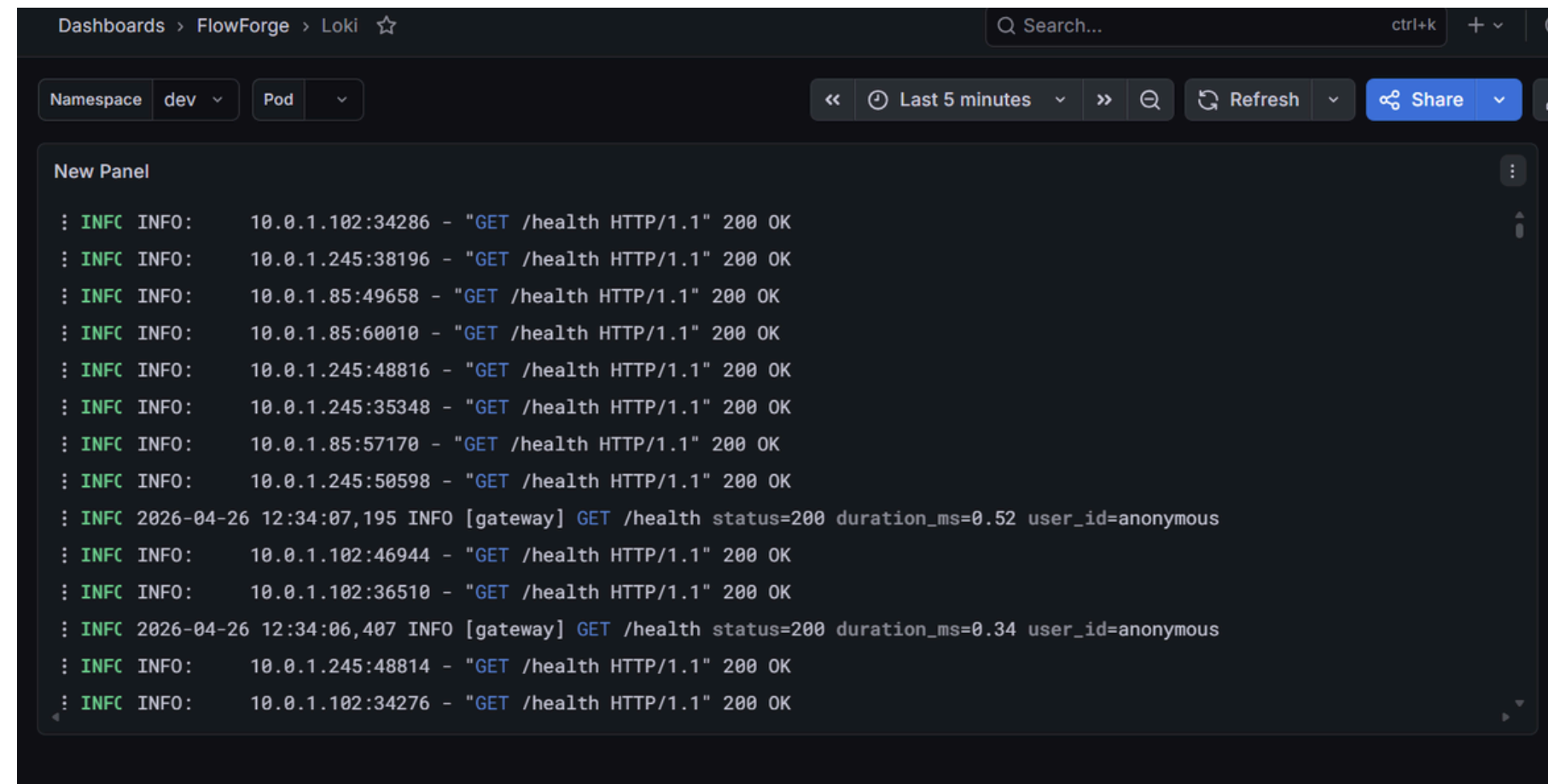
[RESOLVED] KubePodCrashLooping prometheus-kube-state-metrics (redis http 192.168.0.240:8080 kube-state-metrics redis-0 monitoring/prometheus-kube-prometheus-prometheus CrashLoopBackOff warning)

Monitoring



Observability

Loki and Promtail

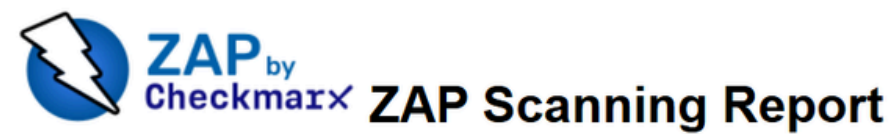


Domain and Subdomains

For all the domains and subdomains SSL/TLS termination is done at HAProxy, all the certificates generated are using certbot

flowforge.fun	Production
dev.flowforge.fun	Testing
preview.flowforge.fun	Preview before going to production
sonar.flowforge.fun	SonarQube Server
argocd.flowforge.fun	ArgoCD Dashboard
rollouts.flowforge.fun	Argo Rollouts Dashboard
headlamp.flowforge.fun	Headlamp
keycloak.flowforge.fun	Keycloak -OIDC
prometheus.flowforge.fun	Alert manager and Prometheus
grafana.flowforge.fun	Grafana Dashboard

DAST



Site: <https://flowforge.fun>

Generated on Sun, 26 Apr 2026 18:52:07

ZAP Version: 2.17.0

ZAP by [Checkmarx](#)

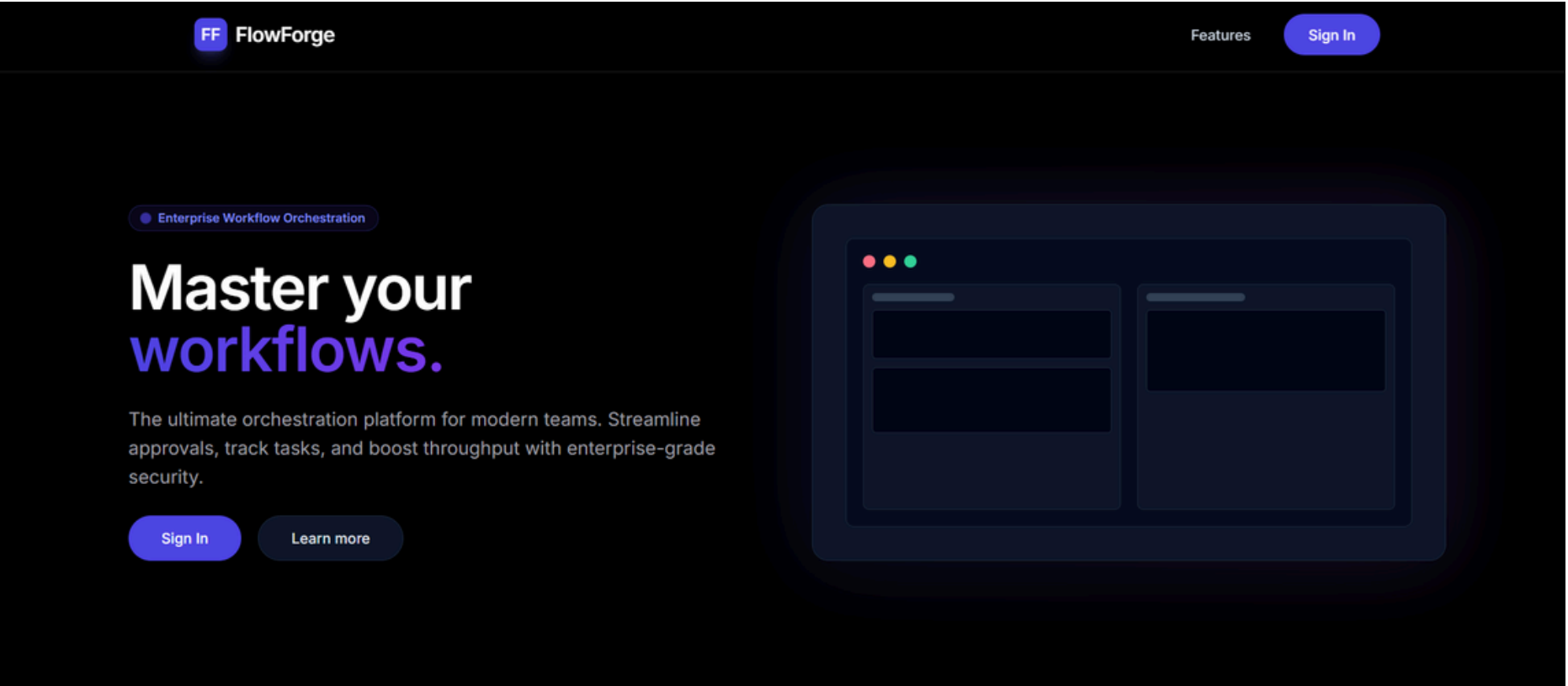
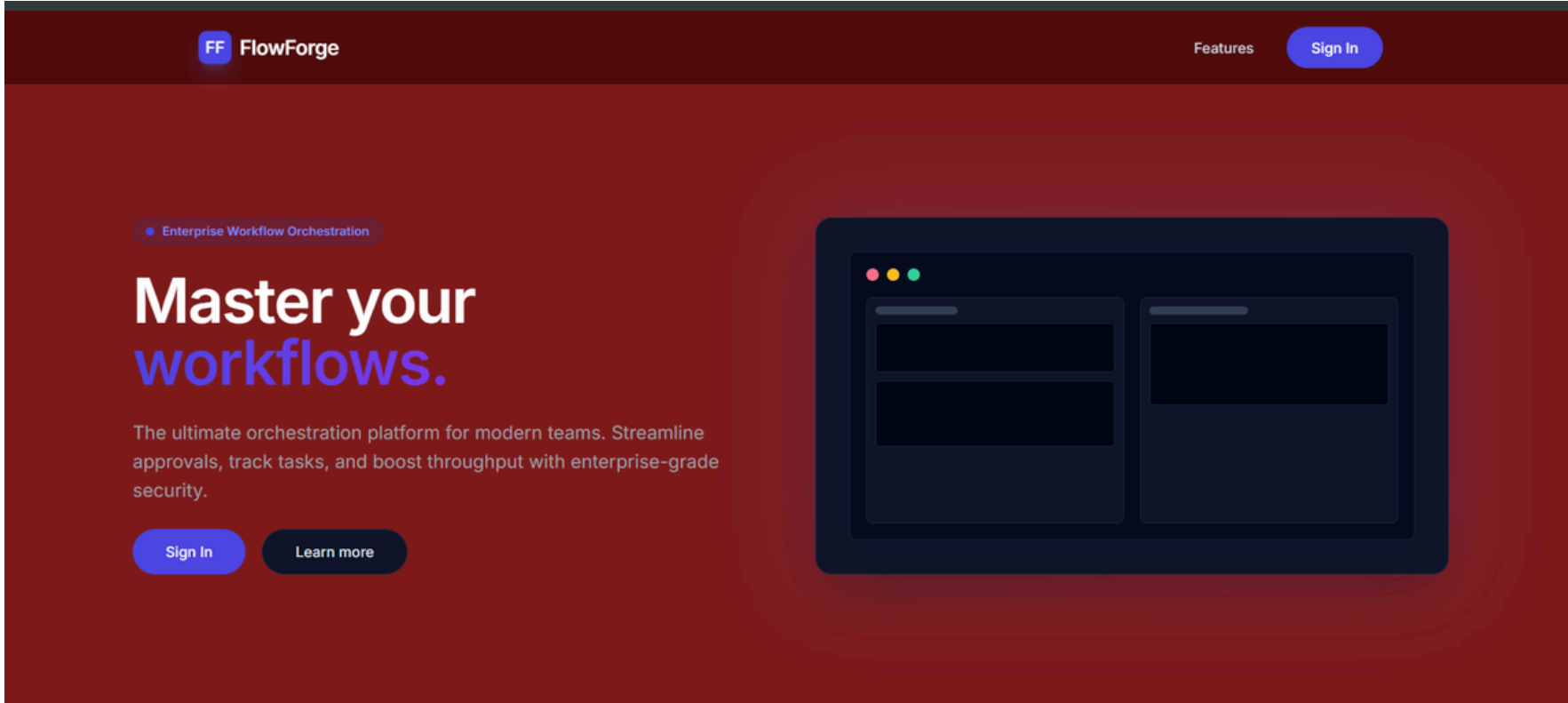
Summary of Alerts

Risk Level	Number of Alerts
High	0
Medium	3
Low	6
Informational	4
False Positives:	0

Insights

Level	Reason	Site	Description	Statistic
Low	Warning		ZAP warnings logged - see the zap.log file for details	5
Low	Exceeded High	https://flowforge.fun	Percentage of responses with status code 4xx	74 %
Info	Informational	http://flowforge.fun	Percentage of responses with status code 3xx	100 %
Info	Informational	https://flowforge.fun	Percentage of responses with status code 2xx	24 %
Info	Informational	https://flowforge.fun	Percentage of endpoints with content type application/javascript	63 %
Info	Informational	https://flowforge.fun	Percentage of endpoints with content type font/woff2	4 %
Info	Informational	https://flowforge.fun	Percentage of endpoints with content type text/css	4 %

Dev vs Prod



Verification

kubectl get all -n prod

```
root@ip-10-0-1-155:/home/ubuntu# kubectl get all -n prod
```

NAME	READY	STATUS	RESTARTS	AGE
pod/flowforge-gateway-846cc4f8cf-99p2c	1/1	Running	2 (34m ago)	20h
pod/flowforge-prod-analysis-service-7f98c86945-8b66t	1/1	Running	1 (33m ago)	7h13m
pod/flowforge-prod-analysis-service-7f98c86945-9k9vr	1/1	Running	1 (35m ago)	7h15m
pod/flowforge-prod-auth-service-59c455cc47-hmn4n	1/1	Running	0	31m
pod/flowforge-prod-auth-service-59c455cc47-n5sxs	1/1	Running	1 (34m ago)	7h23m
pod/flowforge-prod-frontend-765c8845d7-fc5bl	1/1	Running	0	18m
pod/flowforge-prod-frontend-765c8845d7-x9fpf	1/1	Running	1 (34m ago)	7h23m
pod/flowforge-prod-gateway-7f9584ddfd-hhfw7	1/1	Running	1 (33m ago)	7h12m
pod/flowforge-prod-gateway-7f9584ddfd-jmlqg	1/1	Running	1 (35m ago)	7h15m
pod/flowforge-prod-project-service-ffc7bb848-gt8kc	1/1	Running	1 (35m ago)	7h12m
pod/flowforge-prod-project-service-ffc7bb848-sfsfb	1/1	Running	1 (34m ago)	7h15m
pod/flowforge-prod-task-service-76446dc9c-5xlf8	1/1	Running	1 (34m ago)	7h30m
pod/flowforge-prod-task-service-76446dc9c-rpvj2	1/1	Running	1 (35m ago)	7h12m
pod/postgres-0	1/1	Running	1 (35m ago)	8h
pod/redis-0	1/1	Running	2 (33m ago)	21h

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/analysis-service	ClusterIP	10.104.148.100	<none>	80/TCP	25h
service/auth-service	ClusterIP	10.105.153.188	<none>	80/TCP	25h
service/flowforge-gateway	LoadBalancer	10.102.172.177	<pending>	80:30605/TCP	25h
service/flowforge-prod-analysis-service-active	ClusterIP	10.97.243.220	<none>	80/TCP	25h
service/flowforge-prod-analysis-service-preview	ClusterIP	10.109.78.18	<none>	80/TCP	25h
service/flowforge-prod-auth-service-active	ClusterIP	10.103.11.167	<none>	80/TCP	25h
service/flowforge-prod-auth-service-preview	ClusterIP	10.101.171.230	<none>	80/TCP	25h
service/flowforge-prod-frontend-active	ClusterIP	10.101.128.70	<none>	80/TCP	25h
service/flowforge-prod-frontend-preview	ClusterIP	10.108.103.109	<none>	80/TCP	25h
service/flowforge-prod-gateway-active	ClusterIP	10.100.249.87	<none>	80/TCP	25h
service/flowforge-prod-gateway-preview	ClusterIP	10.104.152.101	<none>	80/TCP	25h
service/flowforge-prod-project-service-active	ClusterIP	10.103.190.14	<none>	80/TCP	25h
service/flowforge-prod-project-service-preview	ClusterIP	10.106.74.123	<none>	80/TCP	25h
service/flowforge-prod-task-service-active	ClusterIP	10.99.66.252	<none>	80/TCP	25h

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT (S)	AGE
service/analysis-service	ClusterIP	10.104.148.100	<none>	80/TCP	25h
service/auth-service	ClusterIP	10.105.153.188	<none>	80/TCP	25h
service/flowforge-gateway	LoadBalancer	10.102.172.177	<pending>	80:30605/TCP	25h
service/flowforge-prod-analysis-service-active	ClusterIP	10.97.243.220	<none>	80/TCP	25h
service/flowforge-prod-analysis-service-preview	ClusterIP	10.109.78.18	<none>	80/TCP	25h
service/flowforge-prod-auth-service-active	ClusterIP	10.103.11.167	<none>	80/TCP	25h
service/flowforge-prod-auth-service-preview	ClusterIP	10.101.171.230	<none>	80/TCP	25h
service/flowforge-prod-frontend-active	ClusterIP	10.101.128.70	<none>	80/TCP	25h
service/flowforge-prod-frontend-preview	ClusterIP	10.108.103.109	<none>	80/TCP	25h
service/flowforge-prod-gateway-active	ClusterIP	10.100.249.87	<none>	80/TCP	25h
service/flowforge-prod-gateway-preview	ClusterIP	10.104.152.101	<none>	80/TCP	25h
service/flowforge-prod-project-service-active	ClusterIP	10.103.190.14	<none>	80/TCP	25h
service/flowforge-prod-project-service-preview	ClusterIP	10.106.74.123	<none>	80/TCP	25h
service/flowforge-prod-task-service-active	ClusterIP	10.99.66.252	<none>	80/TCP	25h
service/flowforge-prod-task-service-preview	ClusterIP	10.99.127.96	<none>	80/TCP	25h
service/frontend	ClusterIP	10.103.86.25	<none>	80/TCP	25h
service/gateway	ClusterIP	10.105.45.253	<none>	80/TCP	25h
service/postgres	ClusterIP	None	<none>	5432/TCP	25h
service/project-service	ClusterIP	10.106.254.81	<none>	80/TCP	25h
service/redis	ClusterIP	None	<none>	6379/TCP	25h
service/task-service	ClusterIP	10.110.87.20	<none>	80/TCP	25h

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/flowforge-gateway	1/1	1	1	25h

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/flowforge-gateway-554f8d477	0	0	0	20h
replicaset.apps/flowforge-gateway-5575648476	0	0	0	25h
replicaset.apps/flowforge-gateway-75d85f8bc9	0	0	0	20h
replicaset.apps/flowforge-gateway-846cc4f8cf	1	1	1	20h
replicaset.apps/flowforge-prod-analysis-service-6ddfd6f5bb	0	0	0	8h
replicaset.apps/flowforge-prod-analysis-service-7f98c86945	2	2	2	7h15m
replicaset.apps/flowforge-prod-auth-service-59c455cc47	2	2	2	7h23m
replicaset.apps/flowforge-prod-auth-service-79c985c6cd	0	0	0	8h
replicaset.apps/flowforge-prod-frontend-5978b648dd	0	0	0	25h
replicaset.apps/flowforge-prod-frontend-666dc5d577	0	0	0	25h
replicaset.apps/flowforge-prod-frontend-765c8845d7	2	2	2	7h23m
replicaset.apps/flowforge-prod-frontend-899d46654	0	0	0	21h
replicaset.apps/flowforge-prod-gateway-584cbf99b5	0	0	0	8h
replicaset.apps/flowforge-prod-gateway-7f9584ddfd	2	2	2	7h15m
replicaset.apps/flowforge-prod-project-service-7d6ffbf7f	0	0	0	8h
replicaset.apps/flowforge-prod-project-service-ffc7bb848	2	2	2	7h15m
replicaset.apps/flowforge-prod-task-service-6c9b7f84fb	0	0	0	8h
replicaset.apps/flowforge-prod-task-service-76446dc9c	2	2	2	7h30m

NAME	READY	AGE
statefulset.apps/postgres	1/1	8h
statefulset.apps/redis	1/1	25h

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
horizontalpodautoscaler.autoscaling/flowforge-prod-analysis-service	Rollout/flowforge-prod-analysis-service	cpu: 2%/70%	2	4	2	25h
horizontalpodautoscaler.autoscaling/flowforge-prod-auth-service	Rollout/flowforge-prod-auth-service	cpu: 2%/70%	2	4	2	25h
horizontalpodautoscaler.autoscaling/flowforge-prod-frontend	Rollout/flowforge-prod-frontend	cpu: 1%/70%	2	4	2	25h
horizontalpodautoscaler.autoscaling/flowforge-prod-gateway	Rollout/flowforge-prod-gateway	cpu: 2%/70%	2	4	2	25h
horizontalpodautoscaler.autoscaling/flowforge-prod-project-service	Rollout/flowforge-prod-project-service	cpu: 2%/70%	2	4	2	25h
horizontalpodautoscaler.autoscaling/flowforge-prod-task-service	Rollout/flowforge-prod-task-service	cpu: 2%/70%	2	4	2	25h

root@ip-10-0-1-155:/home/ubuntu#

Conclusion

This project demonstrates a complete, enterprise-ready DevOps platform — from identity management and network segmentation to automated CI/CD pipelines and GitOps-driven deployments. The architecture is secure by default, observable by design, and scalable through pattern-driven configuration. It represents a production-grade implementation of modern cloud-native DevOps principles.